

Environmental variation alters the fitness effects of rifampicin resistance mutations in *Pseudomonas aeruginosa*

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The fitness effects of antibiotic resistance mutations in antibiotic-free conditions play a key role in determining the long-term maintenance of resistance. Although resistance is usually associated with a cost, the impact of environmental variation on the cost of resistance is poorly understood. Here, we test the impact of heterogeneity in temperature and resource availability on the fitness effects of antibiotic resistance using strains of the pathogenic bacterium *Pseudomonas aeruginosa* carrying clinically important rifampicin resistance mutations. Although the rank order of fitness was generally maintained across environments, fitness effects relative to the wild type differed significantly. Changes in temperature had a profound impact on the fitness effects of resistance, whereas changes in carbon substrate had only a weak impact. This suggests that environmental heterogeneity may influence whether the costs of resistance are likely to be ameliorated by second-site compensatory mutations or by reversion to wild-type *rpoB*. Our results highlight the need to consider environmental heterogeneity and genotype-by-environment interactions for fitness in models of resistance evolution.

KEY WORDS: Adaptation, antibiotic resistance, bacteria, fitness.

Studies of antibiotic resistant bacterial isolates have generally revealed that resistance carries a cost, expressed as reduced competitive fitness in the absence of antibiotics (Andersson and Hughes 2010; Vogwill and MacLean 2015). This cost suggests that resistance should be lost from bacterial populations in antibiotic-free environments, such as those that encountered during transmission between hosts (Lipsitch et al. 2000; zur Wiesch et al. 2011). In contrast to this expectation, a large number of in vitro experiments have shown that resistance tends to be maintained in bacterial populations that are propagated in antibiotic-free conditions by selection for compensatory mutations that offset the cost of resistance (Schrag and Perrot 1996; Andersson and Hughes 2010; MacLean and Vogwill 2015). It has been argued that compensatory adaptation occurs more commonly than reversion to antibiotic sensitivity as compensatory mutations outnumber reversion mutations by about an order of magnitude (Björkman et al. 2000; Levin et al. 2000; Poon and Chao 2005; zur Wiesch

et al. 2011). However, such studies have individually focused on only a limited number of environments (e.g., rich laboratory media, murine infection models (Björkman et al. 2000)), and as such it is not known whether compensatory mutations restore fitness across many diverse environments. This is an important point to address, because the environmental specificity of compensatory mutations may play an important role in determining the likelihood of compensatory adaptation as opposed to reversion to antibiotic sensitivity. For example, if compensatory mutations fully recover the cost of resistance mutations across all environments, then environmental heterogeneity is unlikely to impact the long-term maintenance of resistance. On the other hand, it is possible that compensatory mutations only recover the cost of resistance under a subset of environmental conditions. Given that reversion mutations must, by definition, recover the cost of resistance across all environments then environmental heterogeneity may therefore promote reversion.

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Consequently, the role of environmental heterogeneity in promoting compensation or reversion is poorly understood. It is clear that compensatory adaptation is not universal among clinical pathogens (Comas et al. 2012; Farhat et al. 2013)—sensitive isolates tend to have higher competitive ability than co-occurring resistant isolates, and resistance within patients tends to decline following the cessation of antibiotic treatment (MacLean and Vogwill 2015).

Here, we seek to determine whether environmental heterogeneity could constrain adaptation to antibiotic-free conditions in antibiotic resistant strains. Mutations often have environment-specific effects, notably increasing fitness under one set of conditions while decreasing fitness under others (e.g., Maharjan et al. 2013; Carroll et al. 2014). Environmental heterogeneity could therefore alter the propensity for compensatory adaptation or true reversion if the fitness effects of resistance mutations vary across environments. To test this idea, we measured the fitness of three rifampicin resistance mutations across a range of environmental conditions, using the opportunistic pathogen *Pseudomonas aeruginosa*. Rifampicin disrupts bacterial gene expression by binding to a highly conserved domain on the β -subunit of RNA polymerase (*rpoB*), blocking transcript elongation. Resistance evolves by mutations in the rifampicin binding pocket that prevent the drug from binding (Campbell et al. 2001). A frequent route to resistance involves substitutions of a highly conserved histidine residue (*rpoB526H*, using *Escherichia coli* codon numbering). Mutations at this position carry variable fitness costs in different species, including *P. aeruginosa*, *E. coli*, *Salmonella typhimurium*, *Staphylococcus aureus*, and *Mycobacterium tuberculosis* (Andersson and Hughes 2010). Variation at this locus is commonly found in rifampicin-resistant clinical isolates of *M. tuberculosis* following rifampicin treatment (Homolka et al. 2010; Comas et al. 2012; Madania et al. 2012; Ocheretina et al. 2014).

Experiments in *P. aeruginosa* (wild-type *rpoB531H*, using *P. aeruginosa* codon numbering) have shown that populations bearing single nucleotide substitutions at this locus (*rpoB531R* and *rpoB531Y*) carry a cost of resistance in an antibiotic-free, nutrient-rich laboratory culture medium (King's B medium (KB), Hall et al. 2010). Under selection in this environment, the cost of either mutation can be eliminated by acquiring a second nucleotide substitution in the same codon (*rpoB531C*) that is also resistant but inaccessible from wild type (Fig. 1). The wild-type and *rpoB531C* alleles thus represent two fitness peaks separated by the fitness valleys of *rpoB531R* and *rpoB531Y*. The secondary mutation to *rpoB531C* can repair the defects in transcriptional efficiency and gene expression associated with *rpoB531R* and *rpoB531Y* (Qi et al. 2014), making this a classic case of compensatory adaptation. However, it is currently unknown whether *rpoB531C* is compensatory in other environments, or whether *rpoB531R* and *rpoB531Y* are generally costly. We have found

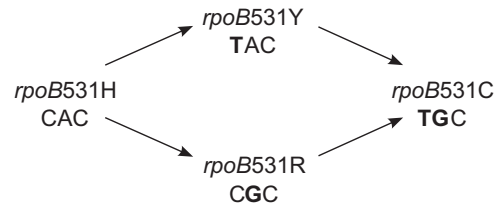


Figure 1. Mutational pathways to *rpoB531C* from *Pseudomonas aeruginosa* wild type.

that the rank order of fitness for the three resistant mutants is generally conserved across environments, meaning that *rpoB531C* is a general compensatory mutation when resistance is costly. However, the fitness effects of all the mutants vary by environment; notably, *rpoB531C* provides no benefit in environments where either *rpoB531R* or *rpoB531Y* carry no cost of resistance. This suggests that environmental heterogeneity may influence the rates of compensatory adaptation and reversion in bacterial populations experiencing a changing environment.

Methods

We used the following *P. aeruginosa* strains: PAO1 (wild type), PAO1::mini-Tn7-GM-*eyfp* (*yfp*-PAO1, gentamycin resistance, and YFP, Choi et al. 2006), and three rifampicin-resistant mutants (*rpoB531R*, *rpoB531Y*, and *rpoB531C*) in a PAO1::mini-Tn7-pLAC-*lux* genetic background (*lux*-PAO1, IPTG-inducible *luxCDABE*, Qi et al. 2014). Whole genome sequencing did not reveal any other mutations (Qi et al. 2014).

To determine the degree of resistance conferred by the *rpoB* mutations, we assayed minimum inhibitory concentration (MIC) of the three rifampicin-resistant strains and the ancestor in 200 μ L of lysogeny broth (LB, Fisher Bioreagents, UK) at 37°C in 96-well microtiter plates. We tested concentrations between 64 mg/L (ancestral MIC) and 1024 mg/L (16 times initial MIC); measuring beyond 1024 mg/L is difficult, as the common solvents used in rifampicin stock solutions (e.g., methanol, used here) begin to affect growth at this concentration.

Competitive fitness of the *rpoB* mutants was measured at three temperatures (30°C, 37°C, and 42°C) in LB and at 37°C in nine of the carbon sources included in the EcoPlate[®] phenotypic microarray (Biolog, Hayward CA, USA) that yielded viable growth (*N*-acetyl-D-glucosamine, L-arginine, pyruvic acid methyl ester, L-serine, 4-hydroxybenzoic acid, polysorbate 80, itaconic acid, D-mannitol, and putrescine). These temperatures were chosen as they are known to have different effects on *P. aeruginosa* growth: 37°C induces expression of virulence genes not expressed at lower temperatures (Wurtzel et al. 2012), and 42°C induces increased expression of the *rpoH* heat shock transcription factor (Potvin et al. 2008). EcoPlates were used as they

provide a breadth of carbon sources favored and disfavored by pseudomonads (and are commonly used for genotype-by-environment interaction studies, e.g., Gravel et al. 2011, Hall 2013). As *rpoB* is a master regulatory gene that has pleiotropic effects on gene expression (Qi et al. 2014), the fitness effects of its mutations are likely to vary over different carbon substrates.

Cultures were inoculated from freezer stocks into 1 mL LB and grown at assay temperature for 24 h, followed by pelleting by centrifugation, diluting 10-fold in M9-broth, and starving (to eliminate residual nutrients) for 24 h at 4°C prior to fitness assays. A 30%/70% mixture of yfp-PAO1 to *rpoB* mutant was created. The mixtures were made by first measuring pure culture population density by flow cytometry (BD Accuri C6; BD Biosciences, San Jose, CA). Events were filtered from noise based on forward scatter (>10,000) and side scatter (>8000, units for both are arbitrary and device-specific). To eliminate “doubles” (i.e., when two cells pass the detector simultaneously), we diluted the pure cultures to ensure event counts were below 1000 events/μL. The population of single cells was then gated on forward scatter and side scatter. For the temperature assays, 200 μL of mixture was added to 800 μL LB ($n = 3$ replicates). For the carbon source assays, 20 μL of mixture was added to 100 μL of M9-broth, and used to rehydrate the EcoPlate® substrates ($n = 9$ replicates). A higher number of replicates were used in the carbon source assays to reduce any potential bias from rare chance beneficial mutations that could have arisen during the competition experiment. Competition cultures were grown for six to seven generations. Initial and final counts of yfp-labeled and unlabeled cells were determined by flow cytometry. To eliminate “doubles,” we diluted the mixed cultures prior to assaying to ensure event counts were below 1000 events/μL. Labeled and unlabeled populations were distinguished by a plot of fluorescence intensity against forward scatter (as larger cells tend to be more fluorescent). Relative fitness of resistant strain (r) over yfp-PAO1 (y) was calculated as $w = \log_2(r_f/r_i)/\log_2(y_f/y_i)$, where i and f indicate initial and final counts. Relative fitness effects of the markers were determined by competing yfp-PAO1 against lux-PAO1 and was used to normalize fitness as $w_{\text{norm}} = w_{\text{resistant}}/w_{\text{yfp-PAO1 vs. lux-PAO1}}$. Fitness data have been uploaded as Supporting Information. Statistical analyses were performed in R 3.0.2 (R Development Core team, 2013).

Results

To evaluate the impact of environmental variation, we first measured competitive fitness of the rifampicin-resistant *P. aeruginosa* strains (*rpoB531R*, *rpoB531Y*, and *rpoB531C*) relative to the sensitive wild-type strain (*rpoB531H*) in nutrient-rich culture medium across a range of temperatures (LB at 30°C, 37°C, and 42°C; Fig. 2A). All *rpoB* mutant strains had a rifampicin MIC that was greater than our detection limit of 1024 mg/L, as compared

with wild-type MIC of 64 mg/L. The rank order of mean fitness was *rpoB531R* < *rpoB531Y* < *rpoB531C* across all temperatures (Tukey's test; $P < 0.001$). However, the fitness effect of each mutation depended on temperature (two-way analysis of variance [ANOVA]: strain: $F_{2,18} = 104$, $P < 0.001$; temperature: $F_{2,18} = 53.1$, $P < 0.001$; interaction: $F_{4,18} = 6.27$, $P = 0.02$). At 37°C, the standard growth temperature for PAO1, *rpoB531C* did not differ in fitness from the wild type (one-sample t -test of estimated mean = 1: $t_2 = -1.5$, $P = 0.27$). At 42°C, *rpoB531C* was significantly less fit than the wild-type *rpoB531H* (estimated mean ≤ 1 : $t_2 = -7.7$, $P = 0.01$); in contrast, at 30°C *rpoB531C* substantially surpassed wild-type fitness (estimated mean ≥ 1 : $t_2 = 16.8$, $P = 0.018$).

As a second test of the effect on environmental variation, we measured competitive fitness across carbon substrates (Fig. 2B; two-way ANOVA: strain: $F_{2,230} = 27.7$, $P < 0.001$; carbon: $F_{8,18} = 8.14$, $P < 0.001$, nonsignificant interaction, $P > 0.05$). The rank-order of mean fitness observed for temperature (*rpoB531R* < *rpoB531Y* < *rpoB531C*) was maintained across environments, except where *rpoB531R* or *rpoB531Y* were cost-free. Although *rpoB531C* was less fit than the PAO1 wild type in four carbon sources, the overall average difference in fitness was not statistically significant (Tukey's HSD: $P = 0.62$). Averaging across all carbon sources, *rpoB531C* had higher fitness than the high-cost mutation (*rpoB531R*; Tukey's HSD: $P < 0.001$), but not the low-cost mutation (*rpoB531Y*; Tukey's HSD: $P = 0.26$). However, *rpoB531C* was more fit than *rpoB531Y* when averaged across carbon sources where *rpoB531Y* was costly (Tukey's HSD: $P = 0.03$). Individually examining each carbon source, *rpoB531C* was statistically more fit than *rpoB531R* on 6/9 carbon sources, but was not more fit than *rpoB531Y* (correcting for false discovery rate [FDR]: critical t -value = 2.53 on degrees of freedom [df] = 16 with $\alpha = 0.05$). We note that some genotype and environment combinations had increased variability in fitness. This was particularly prevalent in environments with the highest fitness costs, consistent with the potential acquisition of adaptive mutations by some replicates during the competition experiment. Although measuring competitive fitness by flow cytometry is usually precise (capable of measuring selection coefficients as small as 10^{-3} , Gallet et al. 2012), precision is reduced when there is significant potential for the acquisition of beneficial mutations. Hence, our data may underestimate the extent of genotype-by-environment interactions.

Discussion

Environmental variation could act as an important constraint on the adaptive trajectories of antibiotic resistant strains if the fitness effects of resistance mutations are environmentally dependent. For a clinically relevant triplet of rifampicin resistance mutations, we found that environment did not generally alter the rank

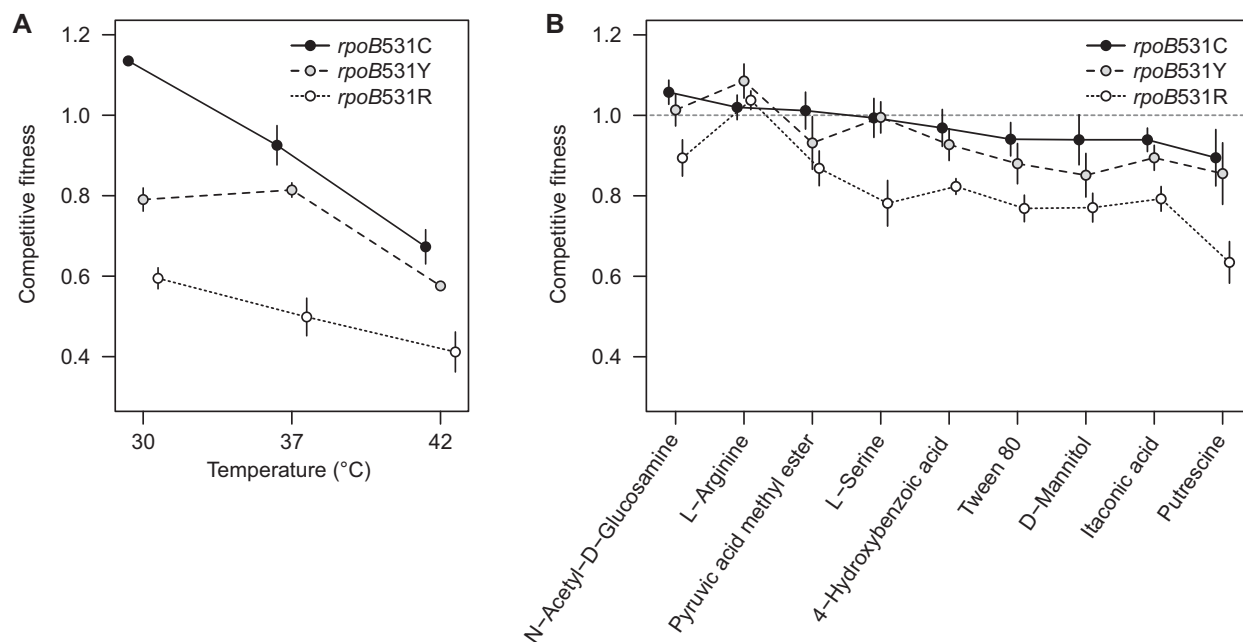


Figure 2. Competitive fitness (means $\pm$ SE) of *Pseudomonas aeruginosa* *rpoB* mutants in varying environments: (A) temperature of LB medium, $n = 3$; (B) carbon source in M9-broth at 37°C, $n = 9$.

order of fitness of the resistance mutations relative to each other ($rpoB531R < rpoB531Y < rpoB531C$), except where resistance was cost-free for either the R or Y allele. This suggests that the second-step rifampicin resistance mutation *rpoB531C* is a general compensatory mutation for both *rpoB531R* and *rpoB531Y*. However, the fitness effects of all three mutations relative to the wild-type *rpoB531H* were contingent on temperature (Fig. 2A), and carbon substrate (Fig. 2B). The fitness of *rpoB531C* was significantly greater than, or less than, the fitness of wild-type *rpoB531H* in some environments. Collectively, these results suggest that environmental heterogeneity is unlikely to entirely restrict the adaptive potential of resistant populations to either compensatory adaptation or true reversion. However, it is possible that exposure to various environmental conditions may increase the frequency of compensatory adaptation in some populations, while decreasing it in others.

Our results suggest that once the *rpoB531C* mutation has become fixed in a population, the population is unlikely to ever revert back to rifampicin-sensitive *rpoB531H*. Due to the structure of the genetic code, H cannot be substituted from C by a single mutational step without first substituting either R or Y (Fig. 1). Combined with the well-conserved rank order of fitness, this implies a rugged fitness landscape over which it is generally impossible to move from a compensated antibiotic resistant peak (*rpoB531C*) to an antibiotic sensitive peak (*rpoB531H*) without passing through an antibiotic resistant fitness valley (*rpoB531Y* or *rpoB531R*). An exception to this general rule would be to first introduce an *rpoB531C* population to L-arginine, which should select for

rpoB531Y, and then to expose the newly fixed *rpoB531Y* population to 42°C LB or putrescine, where *rpoB531C* confers only a small benefit relative to the benefit of reversion to *rpoB531H*. However, adaptation via noncompensatory beneficial mutations that are specific to each new environment would likely hamper this process. The nature of this adaptive landscape may explain the prevalence of the equivalent *rpoB445C* allele among clinical *M. tuberculosis* isolates (Homolka et al. 2010; Comas et al. 2012; Madania et al. 2012), as *rpoB445R* and *rpoB445Y* are also costly (Gagneux et al. 2006). This is consistent with classical predictions that compensatory adaptation should prevent subsequent reversion to sensitivity due to epistasis between compensatory and resistance mutations (Schrag et al. 1997; Weinreich et al. 2005). This idea derives from model systems where second-site compensatory mutations that are costly by themselves offset the cost of resistance via epistatic interactions with resistance mutations. Our system differs slightly from this classical model, as resistance and compensation occur in the same codon, and a reversion of either “resistance” mutation in *rpoB531C* results in a genotype that is still resistant.

A striking feature of our results is that fitness responded only weakly to carbon substrate, but strongly to temperature. Large shifts in gene expression with changing temperature are relatively common in bacteria, particularly as a regulator of virulence-associated genes (e.g., Konkelt and Tilly 2000; Wurtzel et al. 2012), so it is logically consistent that the fitness effects of mutations in a master regulator of gene expression (*rpoB*) would be contingent on temperature. Interestingly, some *rpoB* mutations

are beneficial at 42°C (Rodríguez-Verdugo et al. 2013). Weak genotype-by-environment interactions with carbon substrate, and strong interactions with temperature, are consistent with previous results (e.g., Ostrowski et al. 2005; Laine et al. 2008; Bataillon et al. 2011). However, it remains unclear whether environmental specificity is a general feature of mutations, or if it manifests only under certain environmental stresses.

Conclusions

Our results suggest that environmental heterogeneity could alter the relative frequency of occurrence of reversion and compensation, but not entirely restrict the adaptation of resistant populations to one route or the other. Once compensation has occurred, environmental heterogeneity is unlikely to promote reversion to wild type, except under exacting circumstances. Environmental heterogeneity is particularly relevant to resistance evolution, as the migration of resistance between locales is becoming increasingly apparent. Such variability should be accounted for when predicting the likelihood of reversion or compensation.

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